

Tableau Basics

QUESTION 1

Explain what Tableau is and why it is preferred over traditional spreadsheet tools for data. Also mention any two industries where Tableau is commonly used and explain one use case.

ANSWER

Tableau is a powerful data visualization and business intelligence (BI) tool used to analyze, explore, and present data in an interactive and visual way.				
It helps users turn raw data into dashboards, charts, graphs, and reports that are easy to understand.				
Instead of writing complex code, Tableau uses a drag-and-drop interface, making it user-friendly for both beginners and professionals.				

Why Tableau is Preferred Over Traditional Spreadsheet Tools

1) Better Data Visualization

Tableau creates interactive dashboards with real-time insights

Excel mostly uses static charts

2. Handles Large Data Efficiently

Tableau can process millions of rows quickly

Excel has limitations with very large datasets

3. Real-Time Data Connectivity

Tableau connects directly to databases, cloud, and live data sources

Excel usually works with imported/static data

4. Advanced Analytics

Tableau supports forecasting, trend analysis, and AI insights

Excel requires formulas and manual setup

5. Interactive Dashboards

Users can filter, drill down, and explore data dynamically

Excel dashboards are less interactive

6. Faster Decision Making

Tableau provides instant visual insights

Excel analysis can be time-consuming

Industries Where Tableau is Commonly Used

1. Retail

Used to analyze sales, customer behavior, and inventory trends

Example Use Case (Retail Industry)

A retail company uses Tableau to:

Track daily sales across regions

Identify top-selling products

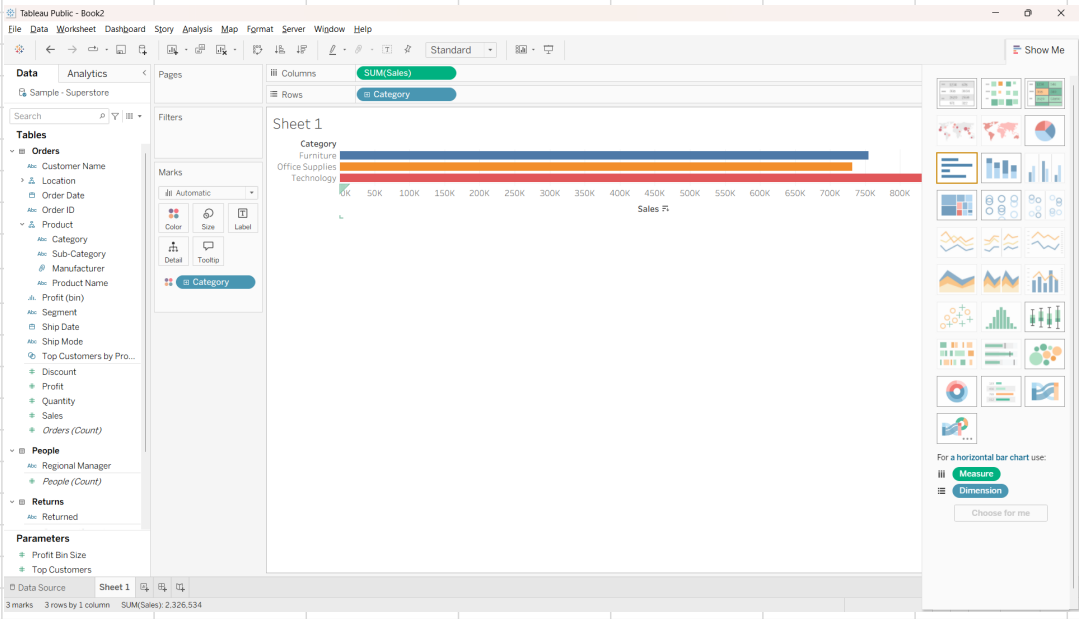
Analyze customer purchase patterns

QUESTION 2	After connecting to the Sample – Superstore dataset: a) List the main types of fields available (examples: time-based, geographical, numerical). b) Identify any four Dimensions and any four Measures from the dataset.								
ANSWER	When you connect to the Sample – Superstore dataset and look at the Data Pane (left side), you'll find different types of fields:								
	1. Categorical Fields (Dimensions)								
	Describe qualitative data								
	Used to group or segment data						b) Four Dimensions and Four Measures		
	Examples: Category, Region, Customer Name								
	2. Numerical Fields (Measures)						In Tableau, fields are divided as:		
	Represent quantitative data						Blue = Dimensions		
	Used for calculations like sum, average						Green = Measures		
	Examples: Sales, Profit, Quantity								
	3. Time-Based Fields (Date/Time)						Four Dimensions (Examples)		
	Used for time analysis						Category		
	Can be broken into Year, Quarter, Month						Sub-Category		
	Examples: Order Date, Ship Date						Customer Name		
							Region		
	4. Geographical Fields						Four Measures (Examples)		
	Used for map visualizations						Sales		
	Automatically recognized by Tableau						Profit		
							Quantity		
	Examples: Country, State, City, Postal Code						Discount		

QUESTION 3

Create a Bar Chart showing Total Sales by Category.
Which field goes to Rows
Which field goes to Columns
Which aggregation is applied to Sales

ANSWER



1 Drag Category → Rows

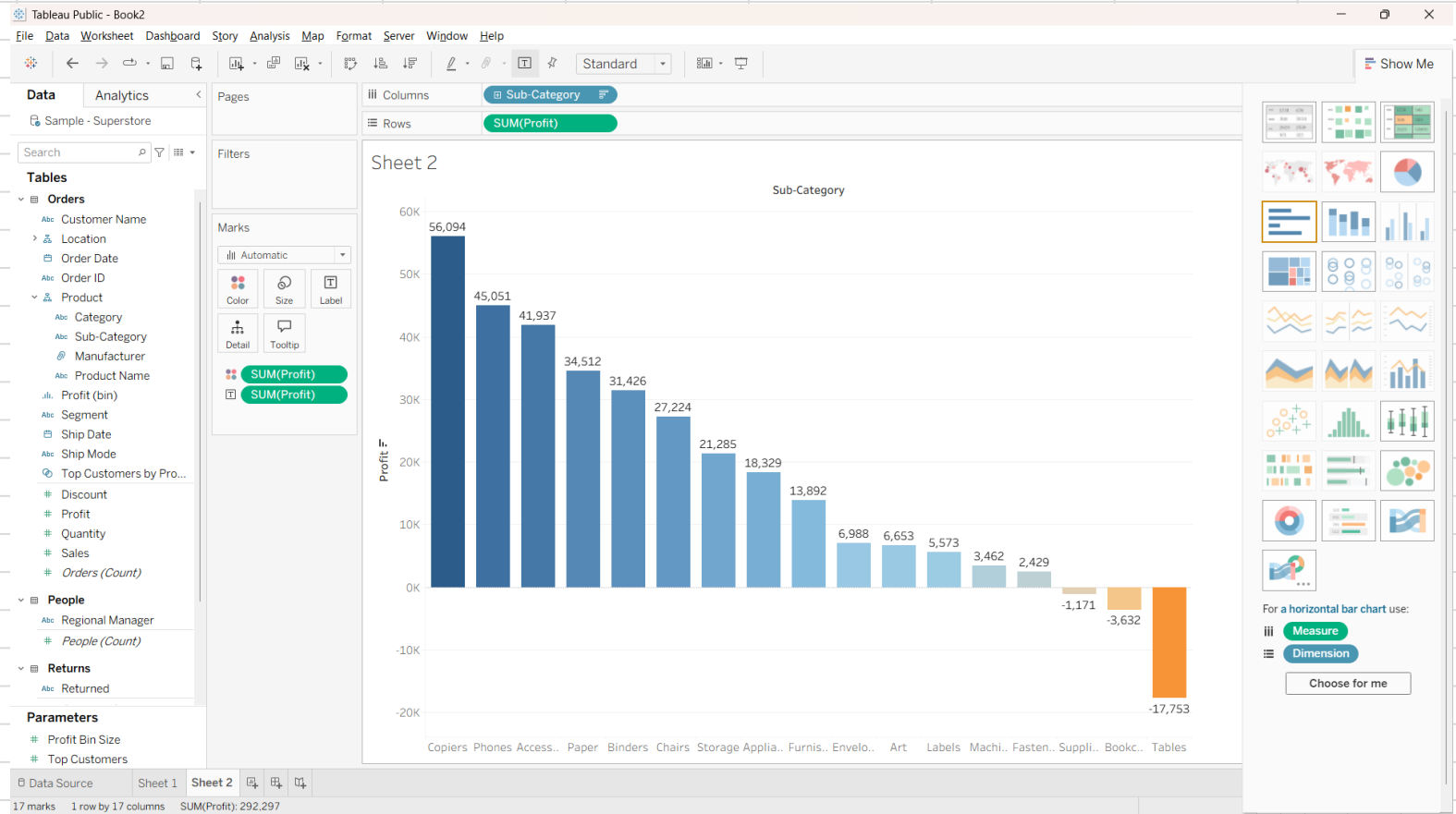
2 Drag Sales → Columns

3 Aggregation: → SUM (Total Sales)

QUESTION 4

What insight can be gained by comparing Sales and Profit together?
Then explain how you would create a chart to analyze Profit by Sub-Category.

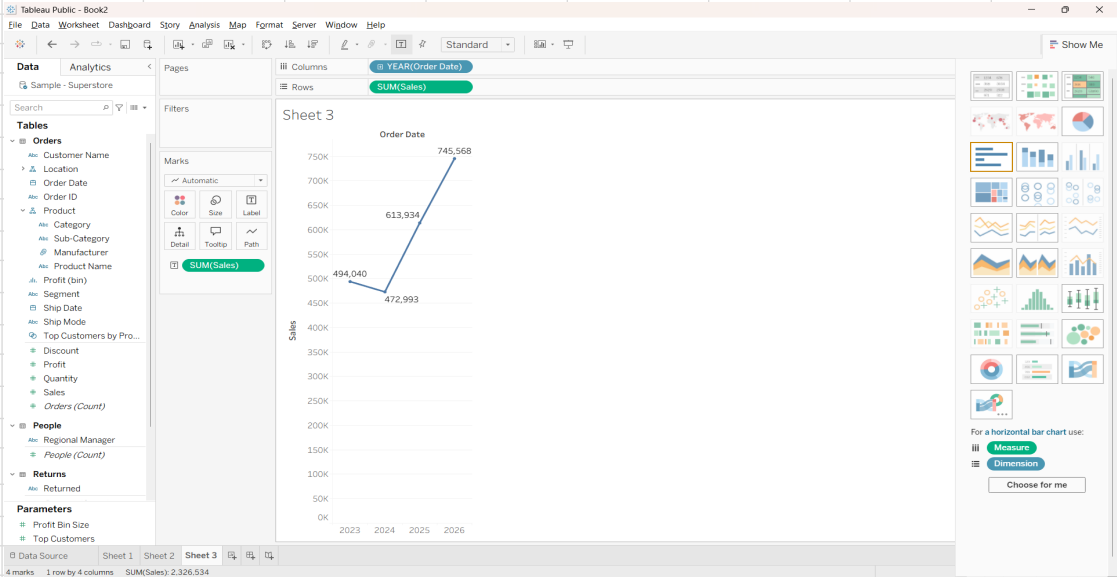
ANSWER



QUESTION 5

Create a Line Chart to show Sales Trend over Order Date.
Which date level would you choose (Year / Quarter / Month)?
Why are line charts suitable for time-based analysis?

ANSWER



Year → Too broad (only shows overall trend, less detail)
Quarter → Moderate detail but still summarized
Month → Best balance of detail and readability

Why Line Charts Are Best for Time-Based Analysis

Shows Trends Over Time
Clearly displays increase or decrease in Sales

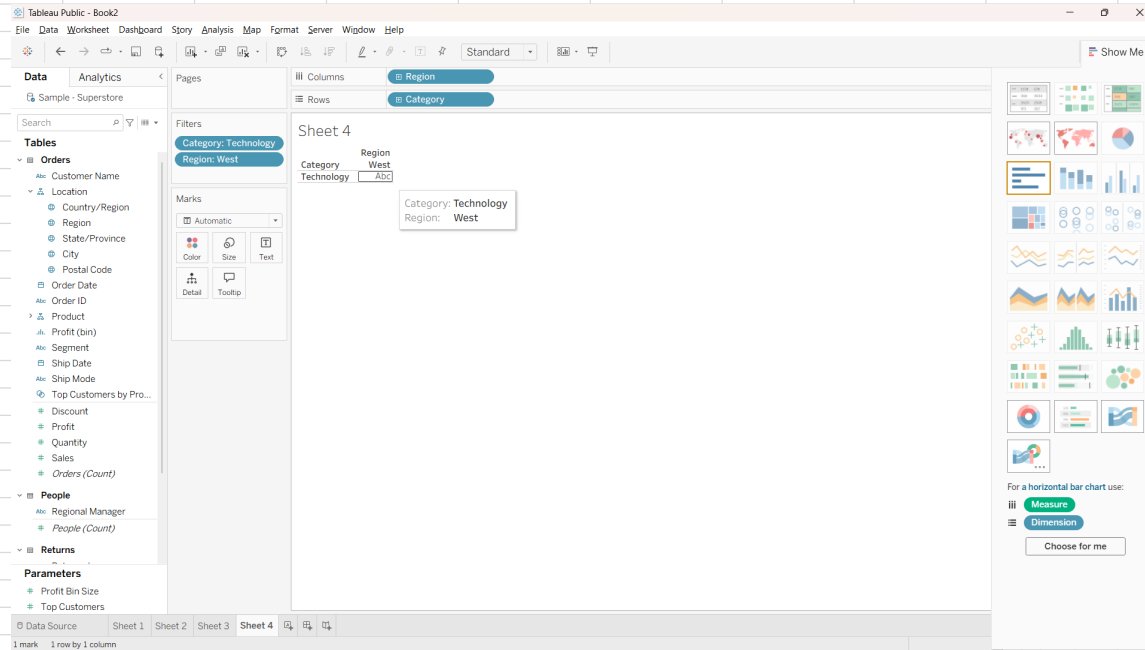
Easy to Identify Patterns
Repeated trends over months or years

Highlights Peaks and Drops
Highest sales periods

QUESTION 6

Apply a Basic Filter to display data only for:
Region = “West”
Category = “Technology”
Explain the steps and mention which shelf is used.

ANSWER



Filters Shelf

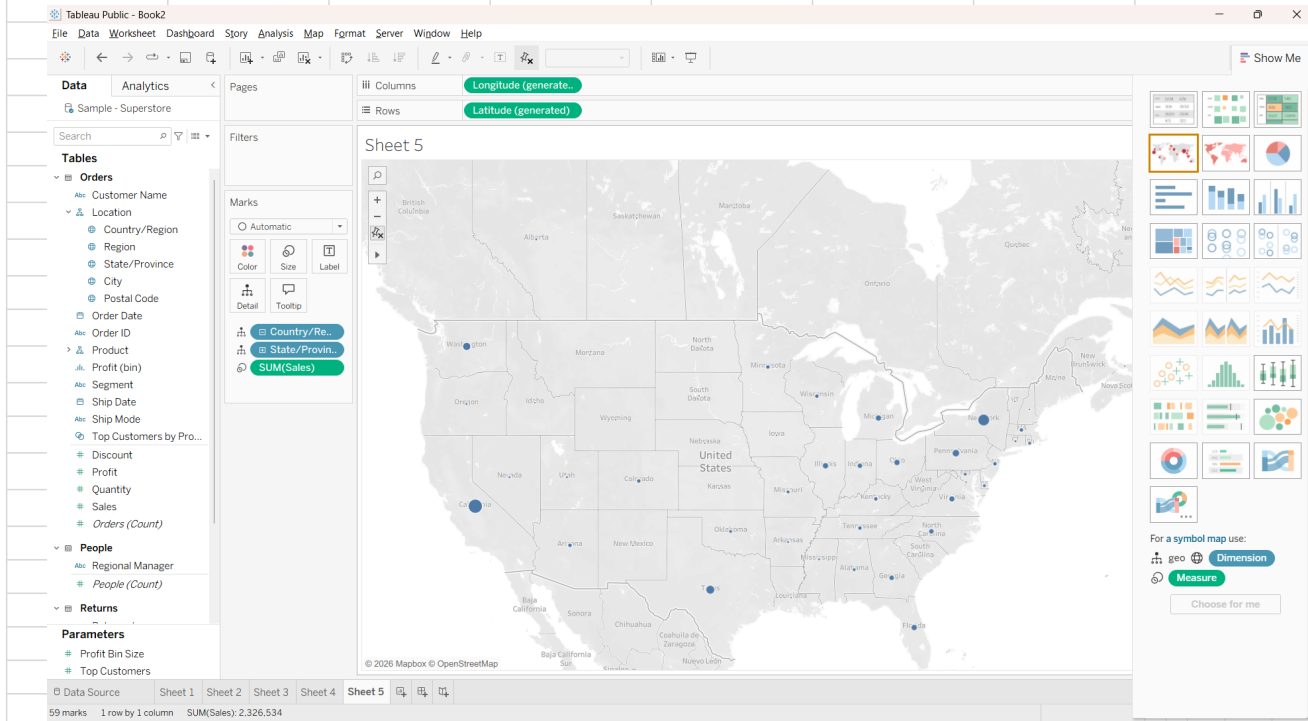
This shelf is used to control which data is displayed in your visualization
It removes unwanted data and keeps only relevant values

QUESTION 7

Explain the business importance of geographical analysis.

Then describe how you would create a Map showing Sales by State using Sample – Superstore.

ANSWER



Business Importance of Geographical Analysis

Identify High-Performing Regions

Find which states or regions generate the most sales

Helps focus marketing and resources

Detect Underperforming Areas

Spot regions with low sales or losses

Helps in improving strategy or operations

Better Resource Allocation

Decide where to invest more (ads, inventory, staff)

Understand Customer Behavior by Location

Different regions have different preferences

Helps in product customization

QUESTION 8		What is the difference between a Worksheet and a Dashboard in Tableau?					
		Why is a dashboard more useful for decision-makers?					
ANSWER		Difference Between Worksheet and Dashboard					
		1. Worksheet					
		A Worksheet is a single view or chart where you create and analyze data.					
		Contains one visualization (bar chart, line chart, map, etc.)					
		Used for detailed analysis					
		You drag fields (Rows, Columns, Filters) to build charts					
		Example:					
		A bar chart showing Sales by Category					
		2. Dashboard					
		A Dashboard is a collection of multiple worksheets displayed together on one screen.					
		Combines multiple charts					
		Can include filters, images, and text					
		Supports interactivity (click, filter, drill-down)					

		Example:					
		Sales by Category					
		Profit by Region					
		Monthly Sales Trend					
		Why Dashboards Are More Useful for Decision-Makers					
		All Insights in One Place					
		No need to switch between multiple sheets					
		Saves time					
		Better Decision Making					
		Combines different metrics (Sales, Profit, Trend)					
		Gives complete business view					
		Interactive Exploration					
		Decision-makers can:					
		Apply filters					
		Click to drill down					
		Storytelling with Data					
		Visual layout makes insights easy to understand					

QUESTION 9		What is Tableau Prep, and why is it important in real-world data projects?				
		Then explain how you would:				
		Remove null values				
		Remove duplicate records				
		Change incorrect data types (Answer in steps, no tool access required.)				
ANSWER		Tableau Prep is a data preparation and cleaning tool used to				
		Clean messy data				
		Combine data from multiple sources				
		Transform data into a usable format				
		It works with a simple drag-and-drop flow, similar to Tableau				
		Why Tableau Prep is Important in Real-World Projects				
		Data is Never Clean				
		Real data has:				
		Missing values				
		Duplicates				
		Errors				
		Saves Time				
		Automates cleaning instead of doing it manually in Excel or SQL				
		Improves Accuracy				
		Clean data = correct insights and decisions				

		Handles Complex Data				
		Combine multiple files (Excel, database, CSV) easily				
		Remove Null Values				
		Open your dataset in Tableau Prep				
		Go to a Clean Step				
		Select the column containing null values				
		Click the filter icon				
		Choose:				
		Remove Duplicate Records				
		Add a Clean Step				
		Identify duplicate rows (same values across columns)				
		Change Incorrect Data Types				
		Go to the column in a Clean Step				
		Click the data type icon (ABC, #, calendar symbol)				
		Select correct type:				
		Text → String				
		Number → Integer/Decimal				
		Date → Date				

QUESTION 10	A retail manager wants answers to the following: Which category generates the highest profit? Which region performs poorly? How does sales change over time? Mention which charts you would use Identify one key insight that management could derive Explain how this helps in business decision-making
ANSWER	Which Charts to Use 1. Highest Profit by Category Bar Chart Category → Rows Profit → Columns Easy to compare profit across categories 2. Poorly Performing Region Bar Chart or Map Region → Rows Profit → Columns 3. Sales Change Over Time Line Chart Order Date (Month) → Columns Sales → Rows

		Best for trend analysis								
		One Key Insight (Example)								
		Technology category generates the highest profit, while the Central region shows low or negative profit, and sales peak during year-end months								
		How This Helps in Business Decision-Making								
		1 . Focus on Profitable Categories								
		Invest more in Technology products								
		Increase marketing and stock								
		2. Improve Weak Regions								
		Analyze why Central region is underperforming								
		Take actions:								
		Adjust pricing								
		Reduce costs								
		Improve distribution								
		3. Plan Based on Sales Trends								
		Identify peak months (e.g., festivals, year-end)								
		Increase inventory and promotions during high-demand periods								
		4. Better Resource Allocation								
		Allocate budget, staff, and inventory based on performance								